MiniPile training dynamics across model sizes

$q_t = \eta\hat{\kappa}_t = 1 - \rho_t \|g_t\|_2 / \|g_{t-1}\|_2$ | unsmoothed; seed 1235

Pythia-14M | Run017

Pythia-31M | Run018

Pythia-70M | Run019

Training loss

$\eta\hat{\kappa}_t$ (proxy)

Raw adjacent cosine ρ_t

Raw gradient $\|g_t\|_2$

Optimizer update

All 712 updates; first pair undefined. Diamonds: partial final batch (311 vs. 1,024 sequences).

AdamW + changing minibatches: q is a gradient-change proxy; it does not establish curvature or stability.

Dashed references: q = 1 and rho = 0. Shared Y scales within rows; gradient norms use log Y.